

Producer Theory

Lecture 11: Companies — Profit Maximization

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Recap: Lecture 10

What we covered last time:

- **Short run** vs **Long run**: fixed inputs vs all variable
- **Cost types**: FC, VC, TC and their per-unit versions (AFC, AVC, ATC)
- **Marginal cost (MC)**: the extra cost of one more unit — the key curve
- **MC crosses AVC and ATC at their minimums** (the “exam grade” rule)
- **LRAC**: the long-run envelope — economies and diseconomies of scale

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 **Today**: Now that we know costs, how does a firm decide **how much to produce** to make the most profit?

Perfect Competition

What Kind of Market Are We Studying?

PERFECT COMPETITION

A market structure where:

1. **Many** buyers and many sellers
2. **Identical** (homogeneous) products
3. **Free entry and exit** — firms can enter or leave the market
4. **Perfect information** — everyone knows prices and quality

Result: No single firm can influence the market price. Each firm is a **price taker**.



Close to perfect competition:

- Agricultural products (wheat, rice)
- Basic tourism services in competitive areas (generic hostels, taxis)
- Street food vendors in a tourist market



Not perfect competition:

- Airlines (few carriers, differentiated)
- Luxury hotels (brand, location matter)
- Disney parks (unique product)

The Price Taker

PRICE TAKER

A firm in perfect competition **cannot choose its price**. The market determines the price through supply and demand. The firm just decides **how much to produce** at that price.

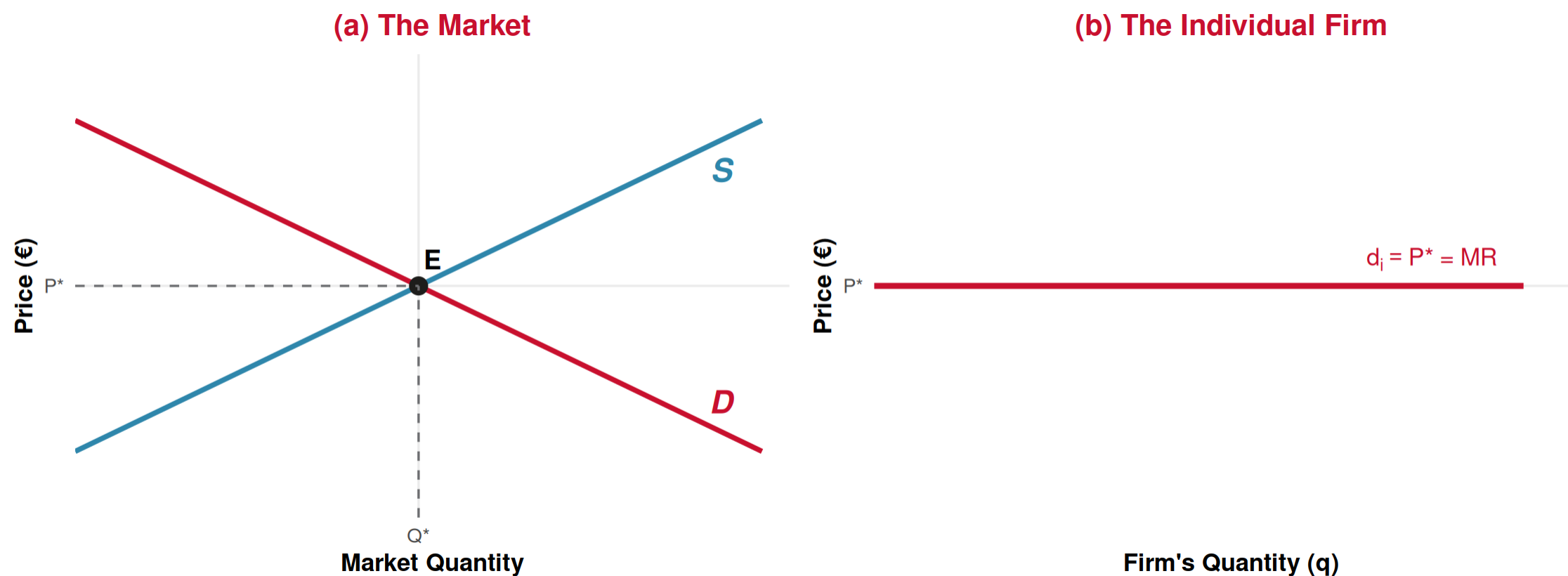
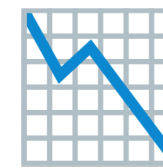
Think of it this way:

- A wheat farmer can't charge €5/kg when the market price is €3/kg — nobody would buy
- There's no reason to charge €2/kg either — they can sell all they want at €3/kg
- So the only decision is: **how many kilograms to produce?**

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Tourism analogy: Imagine dozens of identical beach umbrella rental stands on a long Algarve beach. If one charges €12/day when all others charge €10/day, tourists just walk to the next stand. The market price is €10 — take it or leave it.

The Firm's Demand Curve in Perfect Competition



Left: Market supply and demand determine the equilibrium price P^* .

Right: Each individual firm faces a **perfectly horizontal demand curve** at P^* . It can sell as much as it wants at this price — but **nothing** above it.

👉 For a price taker: **Price = Marginal Revenue**. Every extra unit sold brings in exactly P^* .

The Profit Maximization Problem

Profit: The Basics

PROFIT

$$\pi = TR - TC$$

where $TR = P \times Q$ (total revenue) and $TC = FC + VC$ (total cost).

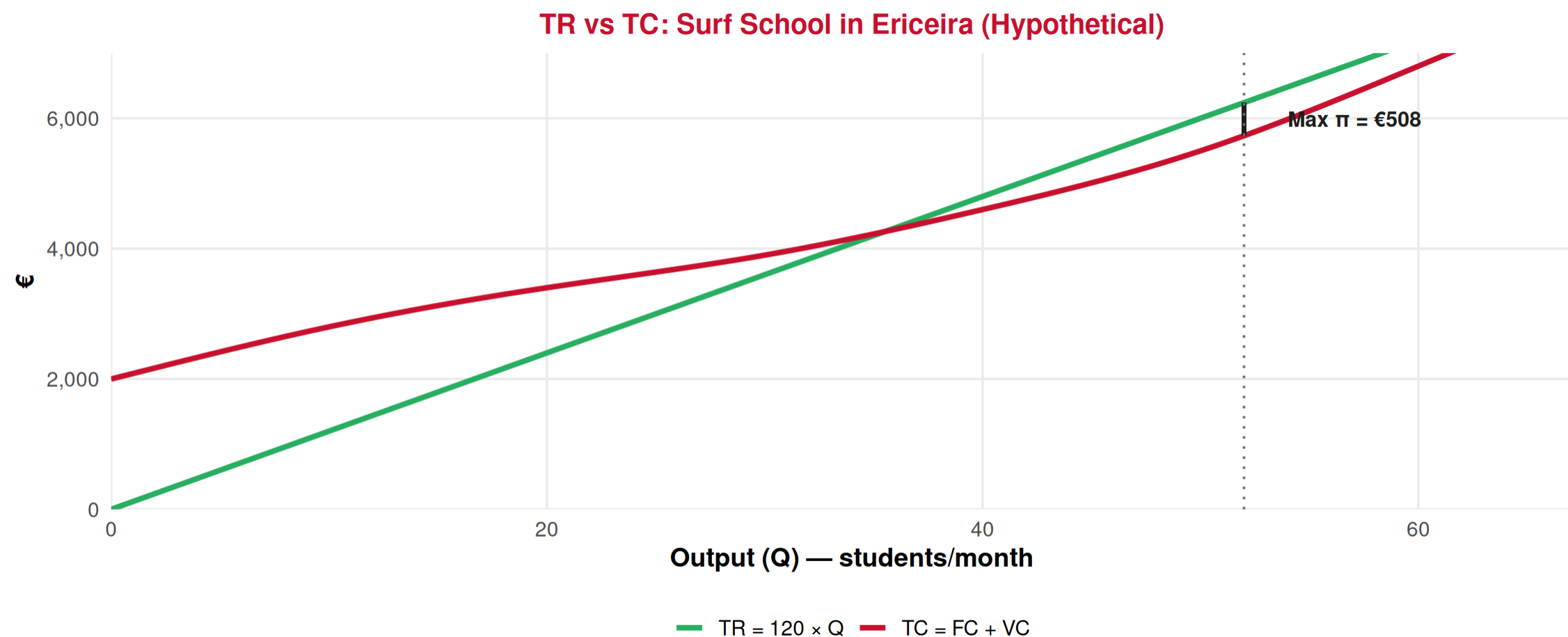
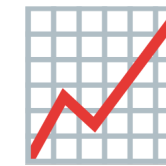
The firm's question: At what quantity Q is $\pi = TR - TC$ as large as possible?

Two equivalent approaches to find the answer:

Approach	Method	Rule
1 TR vs TC	Compare total revenue and total cost curves	Maximize the gap $TR - TC$
2 MR vs MC	Compare marginal revenue and marginal cost	Produce where $MR = MC$

Both give the **same answer**! Approach **2** is more practical and leads us to the supply curve.

Approach 1: TR vs TC — The Total Curves



👉 **Profit** = vertical gap between TR and TC. Maximum profit is where this gap is **widest** — at around $Q^* \approx 52$ students.

This uses the surf school data from last lecture's exercise, with $P = €120/\text{student}$.

Approach 2: MR vs MC — The Per-Unit View

THE PROFIT-MAXIMIZING RULE

Produce the quantity where **Marginal Revenue = Marginal Cost**:

$$MR = MC$$

In **perfect competition**, since $P = MR$, this becomes:

$$\boxed{P = MC}$$

Why does this work? Think about it one unit at a time:

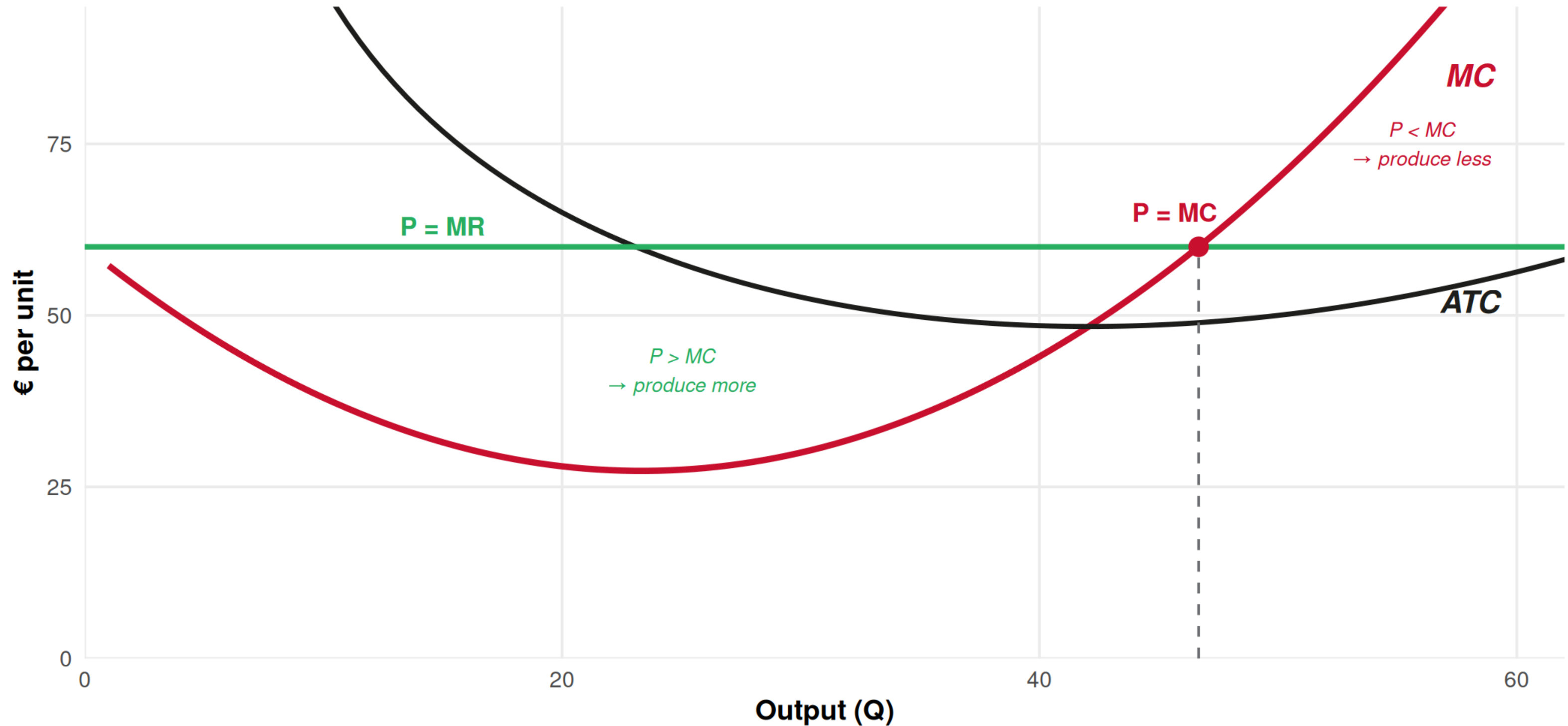
- If $P > MC$ for the next unit → producing it **adds to profit** → produce more! 
- If $P < MC$ for the next unit → producing it **reduces profit** → produce less! 
- If $P = MC$ → you're at the **sweet spot** — no way to increase profit further 



This is just **marginal analysis** from Lecture 3 applied to the firm!

P = MC Graphically

Profit Maximization: $P = MC$ (Hypothetical)



Reading Profit on the Graph

PROFIT AS A RECTANGLE

At the profit-maximizing quantity Q^* :

$$\pi = (P - ATC) \times Q^*$$

This appears on the graph as a **rectangle** with height $(P - ATC)$ and width Q^* .

Three possible outcomes:

-  $P > ATC \rightarrow$ **Positive profit** (green rectangle)
- $\leftrightarrow P = ATC \rightarrow$ **Zero economic profit** (breakeven)
-  $P < ATC \rightarrow$ **Loss** (red rectangle)

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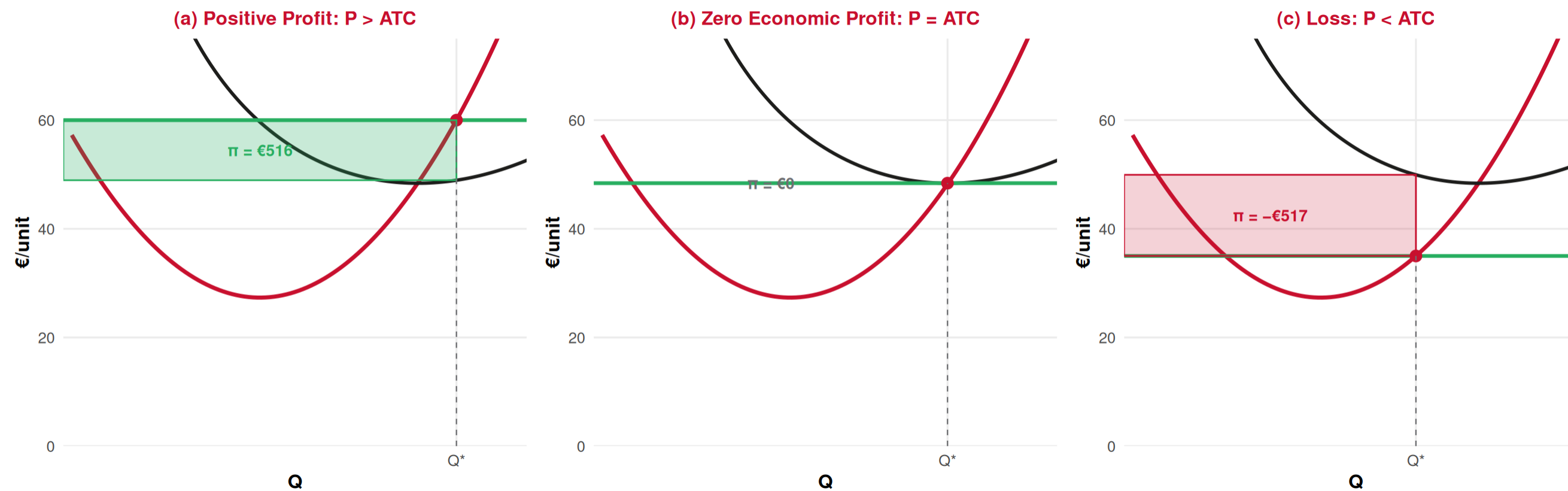
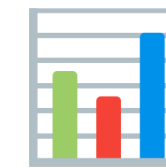
 Next lecture we'll explore the loss case in detail: when should a firm **shut down** vs. **keep operating at a loss**?

Important nuance: “Zero economic profit” is **not bad**!

It means the firm covers **all** costs, including the owner's opportunity cost of time and capital.

The firm earns a **normal return** — the same as its best alternative. No reason to leave!

Profit, Zero Profit, and Loss: Three Scenarios



In all three cases, the firm produces where $P = MC$. What changes is the **relationship between P and ATC** at that quantity.

Worked Example: Tourism

Bottle Factory from the Textbook :bottle:

The sebenta's glass bottle factory has FC = €40/day and the following cost structure:

Q (bottles/day)	Total Revenue (€0.35/bottle)	Labor Cost (€)	TC (€)	Profit (€)	MC (per 100 bottles)
0	0	0	40	-40	—
100	35	10	50	-15	0.10
200	70	20	60	10	0.10
300	105	40	80	25	0.20
400	140	70	110	30	0.30
500	175	110	150	25	0.40
600	210	160	200	10	0.50
700	245	220	260	-15	0.60

Source: Course textbook (sebenta), Table 10

👉 Maximum profit (€30/day) at **Q = 400 bottles/day**. This is where MC is closest to the price (€0.35/bottle) on the rising portion of MC!

What If the Price Changes?

The textbook shows what happens when the bottle price **rises to €0.45/bottle**:

Q (bottles/day)	TR at €0.35	TR at €0.45	TC (€)	Profit at €0.35	Profit at €0.45
0	0	0	40	-40	-40
100	35	45	50	-15	-5
200	70	90	60	10	30
300	105	135	80	25	55
400	140	180	110	30	70
500	175	225	150	25	75
600	210	270	200	10	70
700	245	315	260	-15	55

Source: Course textbook (sebenta), Tables 10 & 11

 **Key insight:** When price rises from €0.35 to €0.45, the firm **produces more** (400 → 500 bottles) and **earns more profit** (€30 → €75). The new optimum is where $MC \approx €0.40$, closer to the new higher price.

The General Lesson



HIGHER PRICE → MORE PRODUCTION → MORE PROFIT

When the market price rises, the $P = MC$ rule tells the firm to expand output along its MC curve. This is the seed of the **supply curve** (Lecture 13).

What higher prices do:

-  Price rises
-  New $P = MC$ intersection at higher Q
-  Revenue rises (both price and quantity)
-  Profit increases

This is why supply curves slope **upward**!

Tourism application:

When hotel room prices rise in Lisbon during peak season:

- Hotels hire more staff, extend hours
- They serve more guests (higher Q)
- Profit per room rises
- More production at higher cost → $P = MC$ at a new, higher quantity

What About Changes in Fixed Costs?

The textbook shows an important result: when **fixed costs rise** (from €40 to €70/day), the **profit-maximizing quantity stays the same** ($Q = 400$).

Why? Because $P = MC$ doesn't involve FC at all! MC only reflects changes in **variable costs**.

FIXED COSTS DON'T AFFECT THE OPTIMAL QUANTITY

FC shifts the total cost curve **up** (and ATC up), reducing profit, but the **optimal Q** stays where $P = MC$.

	FC = €40	FC = €70
Optimal Q	400	400
Profit	€30	€0
MC at Q = 400	€0.30	€0.30

Same Q, but profit drops by exactly the FC increase (€30).

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 FC **does** matter for the **stay or exit** decision — which we cover next lecture!

Marginal Revenue in Detail

Why $P = MR$ in Perfect Competition

Marginal Revenue is the extra revenue from selling **one more unit**:

$$MR = \frac{\Delta TR}{\Delta Q}$$

In **perfect competition**, the firm sells every unit at the same market price P :

$$TR = P \times Q \quad \Rightarrow \quad \text{If } Q \text{ goes up by 1: } \Delta TR = P$$

So $MR = P$ for every unit. That's why the firm's demand curve (d_i) is also its MR curve — a **horizontal line** at P .

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 **This is specific to perfect competition!** A monopolist (single seller) must lower the price to sell more, so $MR < P$. We'll see this in later courses.

For now: **Price taker** $\rightarrow P = MR \rightarrow$ profit-maximizing rule is simply $P = MC$.

The Marginal Approach: Step by Step

Applying the $P = MC$ rule to the bottle factory ($P = €0.35/\text{bottle}$):

point_right: **Stop at $Q = 400$** : the last unit where $MR \geq MC$. Going to 500 would cost more (€0.40) than it earns (€0.35).

Going from...	...to	Extra revenue (MR)	Extra cost (MC)	Profit change	Decision
0 → 100	100	€0.35	€0.10	+€25	✓ Produce
100 → 200	200	€0.35	€0.10	+€25	✓ Produce
200 → 300	300	€0.35	€0.20	+€15	✓ Produce
300 → 400	400	€0.35	€0.30	+€5	✓ Produce
400 → 500	500	€0.35	€0.40	-€5	✗ Stop!

 This step-by-step reasoning confirms the table's answer: **$Q^* = 400$ bottles/day**.

Tourism Application

Hotel Revenue Management: A Profit Maximization Story



Why do hotels charge different prices at different times?

The $P = MC$ rule explains dynamic pricing:

☀️ **Peak season** (summer, Lisbon festivals):

- Market price P is **high** (demand is high)
- $P = MC$ at a **higher Q** → hire more staff, open all rooms
- Profit per room is **large**
- Hotels **expand** services

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👉 The $P = MC$ rule doesn't just say "how much to produce" — it explains the **seasonal rhythm** of tourism businesses!

☁️ **Off-season** (January):

- Market price P is **low**
- $P = MC$ at a **lower Q** → reduce staff, close a wing
- Profit per room is **small** (or even losses)
- Hotels **scale back**

Summary

Today's Key Takeaways:

1. **Perfect competition**: many firms, identical product, price taker
2. **Price taker**: the firm faces a horizontal demand curve at the market price $\rightarrow P = MR$
3. **Profit** $= TR - TC = (P - ATC) \times Q$
4. **Two approaches**: maximize the gap between TR and TC (total curves), or set $MR = MC$ (per-unit curves)
5. **The golden rule**: in perfect competition, produce where $P = MC$ (on the rising portion of MC)
6. If $P > ATC \rightarrow$ positive profit; $P = ATC \rightarrow$ zero economic profit; $P < ATC \rightarrow$ loss
7. **Fixed costs don't affect the optimal Q** — only variable costs (through MC) matter for the production decision
8. **Higher price $\rightarrow$ higher quantity $\rightarrow$ higher profit** — this is the seed of the supply curve

Connection: Lecture 10 gave us the cost curves. Today we used them to find Q^* . Next lecture: what if the firm is making a **loss**? When should it shut down vs. continue?

Next (Lecture 12, March 26): Profits, Shutdown and Breakeven Price Levels. Supply.

Exercises

Practice Time! 

Profit maximization with the $P = MC$ rule.

Exercise 1: Multiple Choice

Question: A small souvenir shop in Sintra operates in a competitive market. The market price for a standard Sintra tile replica is €8. The shop's marginal cost of producing the 50th tile per day is €6, and the marginal cost of the 51st tile is €8.50. How many tiles should the shop produce?

- A. 49 tiles
- B. 50 tiles
- C. 51 tiles
- D. Cannot be determined without knowing fixed costs

Answer: B

At $Q = 50$: $MC = €6 < P = €8 \rightarrow$ producing the 50th tile adds to profit 

At $Q = 51$: $MC = €8.50 > P = €8 \rightarrow$ producing the 51st tile reduces profit 

So produce **50 tiles**. Note: **fixed costs are irrelevant** to this decision (option D is a trap!).

Exercise 2: Multiple Choice

Question: A perfectly competitive firm is producing at Q^* where $P = MC = €25$. At this quantity, $ATC = €20$. What can we say about this firm?

- A. The firm is making a loss and should shut down
- B. The firm is earning zero economic profit
- C. The firm is earning a positive economic profit of €5 per unit
- D. The firm should increase output to reduce ATC further

Answer: C

Profit per unit = $P - ATC = €25 - €20 = €5$. Total profit = $€5 \times Q^* > 0$.

The firm is making **positive economic profit**. It should NOT increase output (D is wrong) because $P = MC$ already — producing more would mean $MC > P$, reducing profit.

Exercise 3: Open Question

A small boat tour operator in Lagos (Algarve) offers coastal cave tours in a competitive market. The market price is **€35 per ticket**. The operator has the following cost structure:

Tours/day (Q)	FC (€)	VC (€)	TC (€)	MC (€/tour)
0	200	0	200	—
1	200	20	220	20
2	200	35	235	15
3	200	55	255	20
4	200	80	280	25
5	200	115	315	35
6	200	165	365	50
7	200	235	435	70

Hypothetical illustrative example

- Calculate TR and profit at each output level. At what quantity is profit maximized?
- Verify that this quantity is consistent with the $P = MC$ rule.
- Calculate ATC at the profit-maximizing quantity. Is the firm earning positive, zero, or negative economic profit?
- Suppose the market price drops to €20. What is the new profit-maximizing quantity? What is profit at this quantity?
- If fixed costs rise from €200 to €300 (e.g., due to a higher boat lease), does the profit-maximizing quantity at $P = €35$ change? Explain.

Exercise 3: Solution — Part a

a) $TR = P \times Q = €35 \times Q$

point_right: Profit is maximized (least negative = smallest loss) at **Q = 4 or Q = 5** (both yield –€140).

Q	TR (€)	TC (€)	Profit = TR – TC (€)
0	0	200	–200
1	35	220	–185
2	70	235	–165
3	105	255	–150
4	140	280	–140
5	175	315	–140
6	210	365	–155
7	245	435	–190

Looking more carefully: at Q = 5, MC = €35 = P exactly, so **Q* = 5** is the profit-maximizing quantity. At Q = 4, the 5th tour still adds €35 revenue at €35 cost — it's worth producing.

⚠ Note: the firm is making a **loss** at every output level! But Q = 5 minimizes that loss. Whether to keep operating or shut down is the topic of the next lecture.

Exercise 3: Solution — Parts b & c

b) Using the $P = MC$ rule:

- At $Q = 4$: $MC = €25 < P = €35 \rightarrow$ produce the 4th tour ✓
- At $Q = 5$: $MC = €35 = P = €35 \rightarrow$ produce the 5th tour ✓ (MR = MC exactly)
- At $Q = 6$: $MC = €50 > P = €35 \rightarrow$ don't produce the 6th tour ✗

So the rule gives $Q^* = 5$, consistent with the table! ✓

c) At $Q^* = 5$:

$$ATC = \frac{TC}{Q} = \frac{€315}{5} = €63 \text{ per tour}$$

Since $P = €35 < ATC = €63$, the firm is earning **negative economic profit** (a loss):

$$\pi = (P - ATC) \times Q = (35 - 63) \times 5 = -€28 \times 5 = -€140$$

The firm is losing €140/day. 

Exercise 3: Solution — Parts d & e

d) At $P = €20$:

- $Q = 1$: $MC = €20 = P \rightarrow$ this is where $P = MC$ on the rising portion
- $Q = 2$: $MC = €15 < P \rightarrow$ but this is the falling portion of MC (careful!)
- $Q = 3$: $MC = €20 = P \rightarrow$ this is on the rising portion 

The profit-maximizing quantity is $Q^* = 3$ (where $P = MC$ on the **rising** portion of MC).

Profit at $Q = 3$: $TR = 20 \times 3 = €60$, $TC = €255$, $\pi = 60 - 255 = -€195$.

The loss worsened from $-€140$ to $-€195$. The lower price hurts!

e) If FC rises from $€200$ to $€300$:

No, the profit-maximizing quantity does not change! FC does not appear in the MC calculation. MC stays the same at every quantity, so $P = MC$ still gives $Q^* = 5$ at $P = €35$.

However, profit falls by exactly $€100$ (the FC increase): $\pi = -€140 - €100 = -€240$.

 Fixed costs affect **how much** profit you earn, but not **how many** units to produce.

Next Lecture

March 26, 2026: Profits, Shutdown and Breakeven Price Levels. Supply. 

If the firm is making a **loss**, when should it keep operating and when should it **shut down**?



Hint: The answer involves comparing P to **AVC**, not ATC!

Thank You!

Questions? 🙋

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Next class: Thursday, March 26, 2026